CSE2023 - HW6

1) a. undirected
no multiple edges
no loops

b. undirected
multiple edges
no loops

c. undirected
multiple edges
3 loops

d. directed
multiple edges
2 loops

e. directed
multiple edges
2 loops

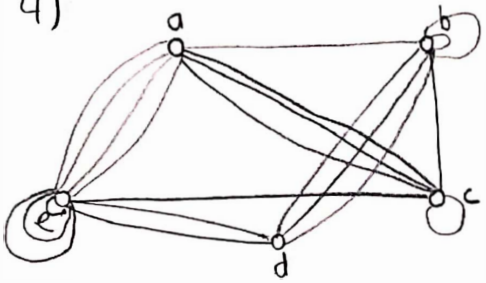
f. undirected
multiple edges
no loops

2) 1. 
2. 
3. 
4. 
5. 
6. 
7. 
8. 
9. 
10. 
11. 
12. 
13. 
14. 
15. 
16. ϕ
17. 
18. 
19. 
20. 
21. 
22. 
23. 
24. 
25. 
26. 
27. 
28. 
29. 
30. 
31. 
32. 
33. 
34. 
35. 

3) Order a, b, c, d

$$\begin{bmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & 1 & 2 \\ 2 & 1 & 1 & 0 \\ 1 & 2 & 0 & 1 \end{bmatrix}$$

4)




5) They are not isomorphic.

6)

| | degree in | degree out | out - in |
|---|---|---|---|
| a | 1 | 2 | 1 |
| b | 3 | 3 | 0 |
| c | 2 | 2 | 0 |
| d | 2 | 2 | 0 |
| e | 4 | 3 | -1 |

There is no euler circuit but euler path exists.

a d e b e d b a e c b c e

7) a, b, d, f are trees.

8)

